Superconducting nanowires at the ePIC far-forward: what a closer approach would buy the polarized-lithium tags

2026-08-26 · companion to plans/09 and to open questions #19 and #20 · computed with `evgen/scripts/nearbeam_aperture_scan.py`, `nearbeam_reach_gain.py` and `nearbeam_sensor_budget.py` on `polligen.reco`, `polligen.recopseudo` and `polli_fastsim.spectator` · primary sources in `refs/`

The far-forward tags of the polarized-lithium programme are beam-blind. An intact ^{6}Li has A/Z = 2 exactly and the α from its breakup has 0.99813 of the beam's rigidity, so neither is separated from the beam by dispersion: what separates them is the transverse *angle* at the interaction point. The acceptance is therefore exp(−B|t|) evaluated at |t| = $(\theta p)^2$ — an exponential in the *square* of the approach angle times the square of the momentum — and approach distance is the whole measurement. Superconducting nanowire detectors, developed for nuclear physics by the Argonne MEP and Physics Divisions, are the candidate technology. This note prices what a closer approach is worth, asks whether a nanowire can deliver it, and separates the part of the case that survives scrutiny from the part that does not.

The verdict, and it is negative. Neither proposed role survives, and both fail on arithmetic rather than on unknowns. (1) **The near-beam role.** A closer approach is worth a great deal — ×26 at 5 × 41, ×569 at 10 × 100, and at mid energy the difference between a fit that returns a_t = 0.48 ± 1.45 and one that returns four clean |t| bins (§2). But the gap is set by 16 mm module tiling and by the 10σ rule, and a nanowire's advantage over slim-edge silicon is its dead edge — worth *0.005 mrad* against a 0.5–3 mrad gap. The 10σ rule is machine protection and halo, not sensor survival, so radiation hardness does not relax it either (§3). (2) **The charge-ID role.** A nanowire latches and delivers *one bit* per plane — and that turns out to be *almost enough*: against the Neyman-Pearson optimum a one-bit, four-plane threshold loses a factor 1.4, while the standard analogue estimator (a truncated mean) loses a factor 86 *to* one bit. Bits are not the issue. What kills the nanowire is **geometric fill factor**: the coincidence that makes one bit sufficient needs every plane to record the track, and at the ANL device's 50% fill a four-plane array cannot reach 95% ^{6}Li efficiency at any working point (§4.1). (3) **Open question #19 has been asked of the wrong technology.** Better still, the α + d breakup that #19 exists to reject is *two hits*, and at beam rigidity they land 6.7 to 45 mm apart — 13 to 90 pixels of the existing 500 μm pitch. Topology, not dE/dx (§4.3).

What does survive, and it is the reason to read on. The acceptance-versus-aperture curve of Figure 1 is technology-independent: it prices *any* closer approach, by any means, and says the mid-energy configuration is worth ×569. And a correction that is independent of nanowires and more immediately actionable than anything else here — **the ePIC pot geometry has moved since the snapshot this repository measured**, in opposite directions at the two ends (§6). Every aperture-conditional number in the programme needs re-measuring.

1 · Why the angle is everything

Both far-forward lithium tags are rigidity-degenerate with the beam. The coherent measurement needs the intact ^{6}Li, whose A/Z is exactly the beam's; the tagged measurements need the α spectator, whose rigidity is 0.99813 of it. Neither is swept aside by the dipoles. The only handle is the transverse angle, and because the coherent recoil follows exp(−B|t|) with B ≈ 50 GeV^{-2} and |t| = $(\theta_x p)^2$, the tagged fraction falls as the exponential of the *square* of the approach angle times the *square* of the beam momentum. Halving the approach angle at 10 × 100 is not worth a factor of two; it is worth a factor of 569.

Two angles compete to set that approach. The **10σ envelope** — 0.727 mrad for the high-acceptance optics — is an operational retraction, a machine-protection rule about beam halo. The **geometric aperture** of the sensor package is what the silicon, its RF shield, its ASIC and its module granularity actually leave open. Shooting an intact ^{6}Li through the ePIC geometry (`tools/fullsim`) measured the second: $|\theta_x| \gtrsim$ 2.00 / 1.35 / 1.03 mrad at 5 × 41 / 10 × 100 / 18 × 275, or 27.5 / 18.6 / 14.2 σ_θ — a package that stops 4.2σ to 17.5σ short of where the machine would allow it. §6 revisits whether that measurement still describes today's geometry; the curves below do not depend on the answer.

2 · What a closer approach is worth

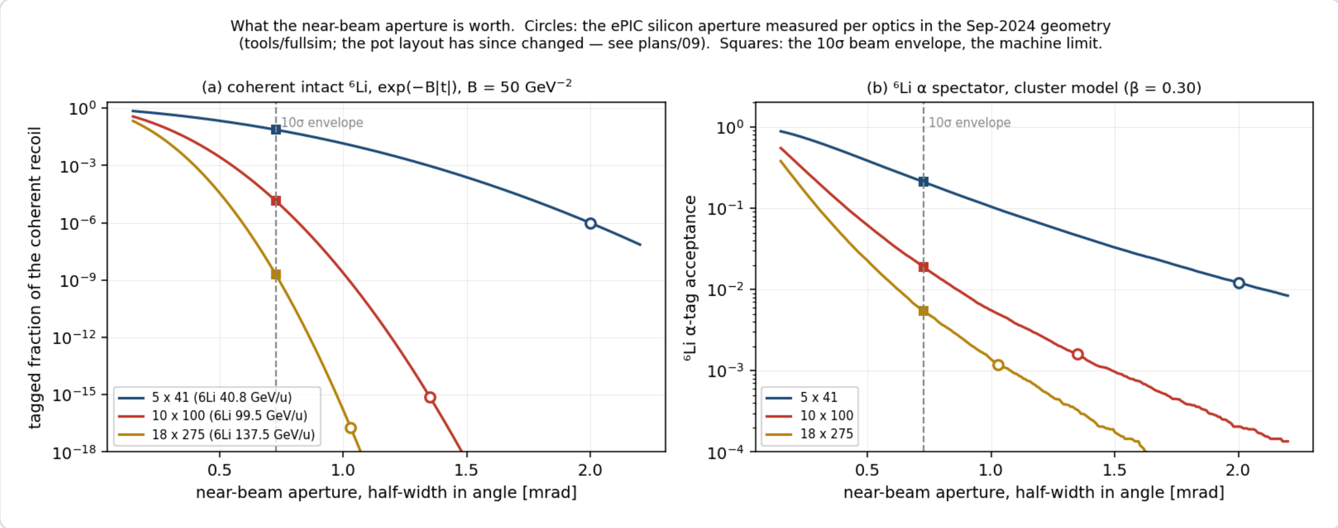

Figure 1 · The price of every aperture. Tagged fraction of the coherent intact ^{6}Li (a) and of the ^{6}Li α spectator (b) against the near-beam half-width in angle, per beam configuration. Circles mark the ePIC silicon aperture measured in the September-2024 geometry; squares mark the 10σ envelope. The vertical half-width is held at its measured value per optics (3.0 / 3.0 / 2.3 mrad). Note the vertical scale of (a): eighteen decades. `nearbeam_aperture_scan.py`.

The analytic gains are large and very unequal:

	5 × 41 (40.8 GEV/U)	10 × 100 (50)	18 × 275 (137.5)
coherent tagged fraction, silicon aperture	1.41×10^{-2}	5.12×10^{-5}	1.94×10^{-17}
coherent tagged fraction, 10σ envelope	3.71×10^{-1}	2.91×10^{-2}	1.97×10^{-9}
gain	**×26**	**×569**	$\times1.0\times10^{8}$, still dead
^{6}Li α tag, silicon aperture	0.103	0.024	0.0012
^{6}Li α tag, 10σ envelope	0.550	0.137	0.0054

Coherent: `reco.rp_hole_acceptance` at B = 50 GeV^{-2} with the measured rectangular cutout. α tag: the cluster model of `polli_fastsim.spectator` at β = 0.30.

The top energy is not rescued, by any technology. At 825 GeV even a 10σ approach demands $p_T > 0.60$ GeV, $|t| > 0.36$ GeV^2, and $\exp(-50 \times 0.36) = 10^{-9}$. No aperture makes the coherent channel live at 18 × 275; the client there is the α tag and nothing else.

2.1 · Through the chain, not just the acceptance

Acceptance is not the measurement. Running the full coherent chain — Roman-Pot emulation, the two azimuths $\alpha = \varphi_e - \varphi_S$ and $\beta = \varphi_t - \varphi_S$, the spin-state-sorted 2-D harmonic fit — at both apertures shows three effects, not one.

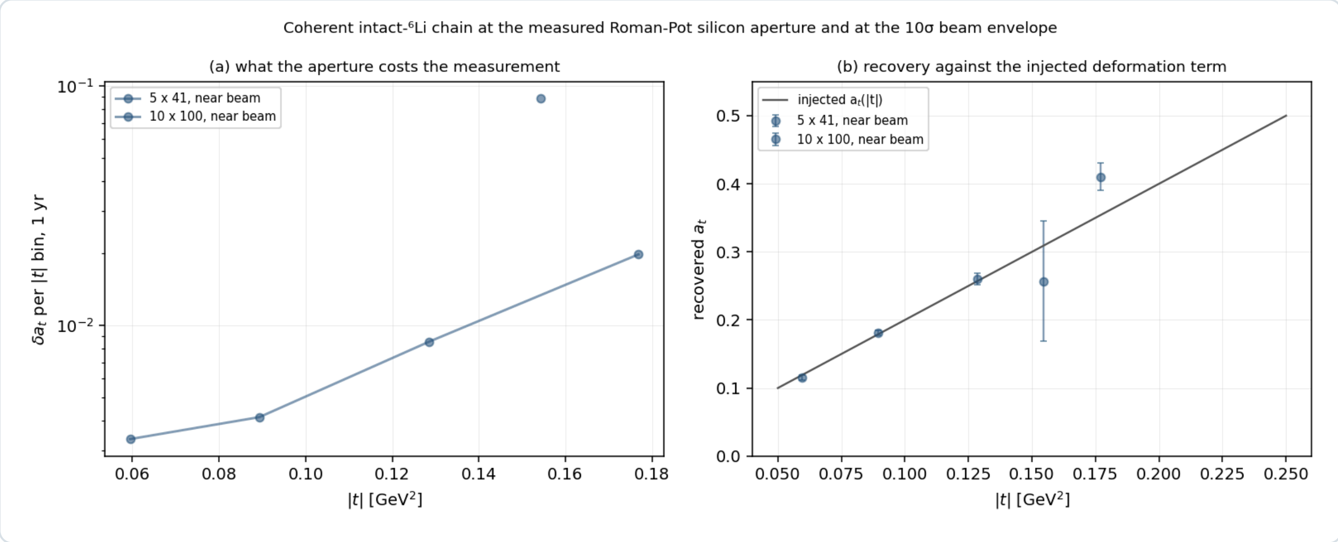

Figure 2 · What the aperture costs the measurement. (a) δa_t per |t| bin at 1 year, at the measured silicon aperture (dashed, squares) and at the 10σ envelope (solid, circles), for the two configurations where the coherent channel is alive. (b) The recovered a_t against the injected deformation term; the off-scale silicon point at 10 × 100 is labelled. `nearbeam_reach_gain.py`.

OPTICS	APERTURE	ACCEPTANCE	|T| BINS	ΔA_T PER BIN, 1 YR	A_E (INJECTED 0.0100)
5 × 41	2.00 mrad, silicon	1.43×10^{-2}	2 of 4	0.046, 0.015	−0.0060 ± 0.0045, +0.0197 ± 0.0044
5 × 41	0.727 mrad, 10σ	3.62×10^{-1}	3 of 4	0.0019, 0.0035, 0.0078	+0.0084 ± 0.0016, +0.0087 ± 0.0031, +0.0012 ± 0.0076
10 × 100	1.35 mrad, silicon	1.02×10^{-4}	2 of 4	1.45, 0.175	−0.068 ± 0.049, −0.019 ± 0.020
10 × 100	0.727 mrad, 10σ	3.25×10^{-2}	4 of 4	0.0069, 0.0044, 0.0077, 0.0168	+0.0117 ± 0.0013 … +0.0304 ± 0.0118

1. **Statistics.** δa_t improves ×8 to ×40 where the measurement already exists.
2. **A configuration.** At 10 × 100 the silicon aperture returns $a_t = 0.48 \pm 1.45$ against a truth of 0.32 — not a measurement. At 10σ it returns four clean bins. The mid-energy configuration goes from dead to the best-covered in the programme, which matters because the reconstruction-chain report otherwise concludes that the coherent channel is a low-energy-only programme. *That conclusion is aperture-conditional, not physics-conditional.*
3. **Bias, not only error.** At 5 × 41 with the silicon aperture the gluon-transversity coefficient comes back −0.0060 ± 0.0045 and +0.0197 ± 0.0044 against an injected 0.0100 — both wrong by more than 2σ, because the emptied azimuthal circle leaves the seven harmonic columns nearly degenerate. At 10σ every bin is consistent. A tight aperture does not merely cost events; it biases the extraction.

A trap this exposed. Money plot 6R carries `cut_scale_x = 2.5` from the pre-measurement belief that the pots surround a *wide horizontal slot*. With the geometric aperture measured, carrying 2.5 as well imposes a 25σ horizontal retraction that no machine requirement asks for — and it binds *before* the geometry does, hiding the entire effect. Both new scripts hold the envelope at 10σ in each axis and let the measured geometry be the only aperture.

3 · The idea is already on record

Near-beam recoil tagging is not a new application to propose to the Argonne group. It is theirs, twice over. The EIC Yellow Report §14.5.3 lists the four SNSPD applications the EIC R&D committee identified, of which the first is *"a Roman pot detector in the forward region about 35 meters or more from the interaction point to tag low momentum transfer recoiling ions"* [1]. Armstrong's FY22 EIC generic-detector-R&D proposal — reviewed and funded — states the mechanism verbatim: *"The so-called 10 σ rule-of-thumb for Roman pots prevents the beam from damaging the detectors by limiting its proximity to the beam. Radiation hard nanowire detectors will be able to move closer to the beam, thus extending the acceptance reach at lower t"* [2].

What this study adds is the number attached to *lower t*, for a species whose acceptance is exponentially more sensitive to it than the DVCS protons that motivate the rest of the far-forward programme. Table 2 is that number.

> **But the stated mechanism does not hold.** The 10σ rule is a machine-protection and beam-halo rule, not a statement about sensor damage: it exists to keep the *beam* safe and the detector out of the halo, and a more radiation-hard sensor does not relax it. LHC ran the CT-PPS pots at 25σ at 6.5 TeV. The ePIC geometry says as much in the file itself — the per-energy insertions are *"the ten-sigma cuts for the Roman pots, translated to the physical layout we currently have"* [3].
>
> **And the fallback argument does not survive arithmetic either.** One could argue granularity and dead edge instead: the insertion is quantised by the module, and a nanowire's active area reaches within a few hundred nanometres of the substrate edge [1] — measured, in the SMSPD case, to be sharper than the 10 μm telescope used to look at it [8]. That is a real and unusual sensor property. It is simply not what limits this measurement: slim-edge planar and 3D silicon already deployed in Roman Pots reaches 100–200 μm of dead edge, so the difference is ≈ 150 μm, or **0.005 mrad at R_{12} = 30.6 m** — 1% of the 0.52 mrad edge and 0.3% of the 10 × 100 gap. Unmeasurable in |t|.
>
> **The granularity half of the argument does survive — and it belongs to ePIC, not to a detector R&D group.** At 10 × 100 the blind block is 32 mm wide against a $10\sigma_x$ of order 10–17 mm: a factor ~330 in coherent yield surrendered to retracting a whole 32 mm-wide block. The fix is a staircase retraction near x = 0, narrower modules, or the x-moving layer ePIC is *already designing* — Jentsch, July 2025: *"Simplest option would be one 'x' moving layer and one 'y' moving layer in each station … Expect to have this finalized in the next 6-8 weeks."* This is the highest-value item in the study, and it is a layout change to a baselined detector, not new technology.

4 · Charge identification: the mechanism is right, the technology is not

Open question #19 has no answer at IP6. An intact ^{6}Li, an α and a d from breakup share rigidity *and* velocity; only dE/dx separates them, going as z^2 — 9 : 4 : 1 — and the EICROC AC-LGAD chain records pulse amplitude for charge-sharing but no EIC document addresses Z-ID in the pots. The only documented concept is a z^2 Cherenkov behind the IR-8 secondary-focus pots.

A nanowire will not do it by pulse height. The device latches: the pulse amplitude is the diverted *bias* current, and both Fermilab/JPL beam papers show the amplitude distributions from 120 GeV hadrons, 120 GeV muons, 8 GeV pions and 8 GeV showering electrons to be indistinguishable, with one amplitude threshold serving all [7,8]. Any scheme that reads Z off a pulse height is dead on arrival.

It would do it by firing threshold. In the normal-core hot-spot regime a deposit Q makes a normal core of radius $r_s \propto \sqrt{Q}$, and the wire switches when that core spans it against the bias-suppressed superconducting width:

$$r_s = \sqrt{\frac{Q}{e\pi c\rho(T_c - T_0)}}\,, \qquad \frac{I_{\rm th}}{I_c} = 1 - \frac{2r_s}{w}$$

Three groups converge on this relation from three directions — Argonne's own Eqs. 1–2 for 120 GeV protons [4], Renema's $E = (w/C)^2(1 - I_b/I_c)^2$ for photons, and the ion literature's $I_{th}/I_c = 1 - (zeV)^{1/2}C/w$ [9]. Because dE/dx ∝ z^2 at fixed velocity, and because a ^{6}Li at 137.5 GeV/u has βγ = 148 against 128 for Argonne's calibration proton — the same velocity to 15% — **r_s scales as z linearly, with no velocity correction**: 134 nm (p, d), 268 nm (α), 402 nm (^{6}Li), anchored on Argonne's own measured 134 nm.

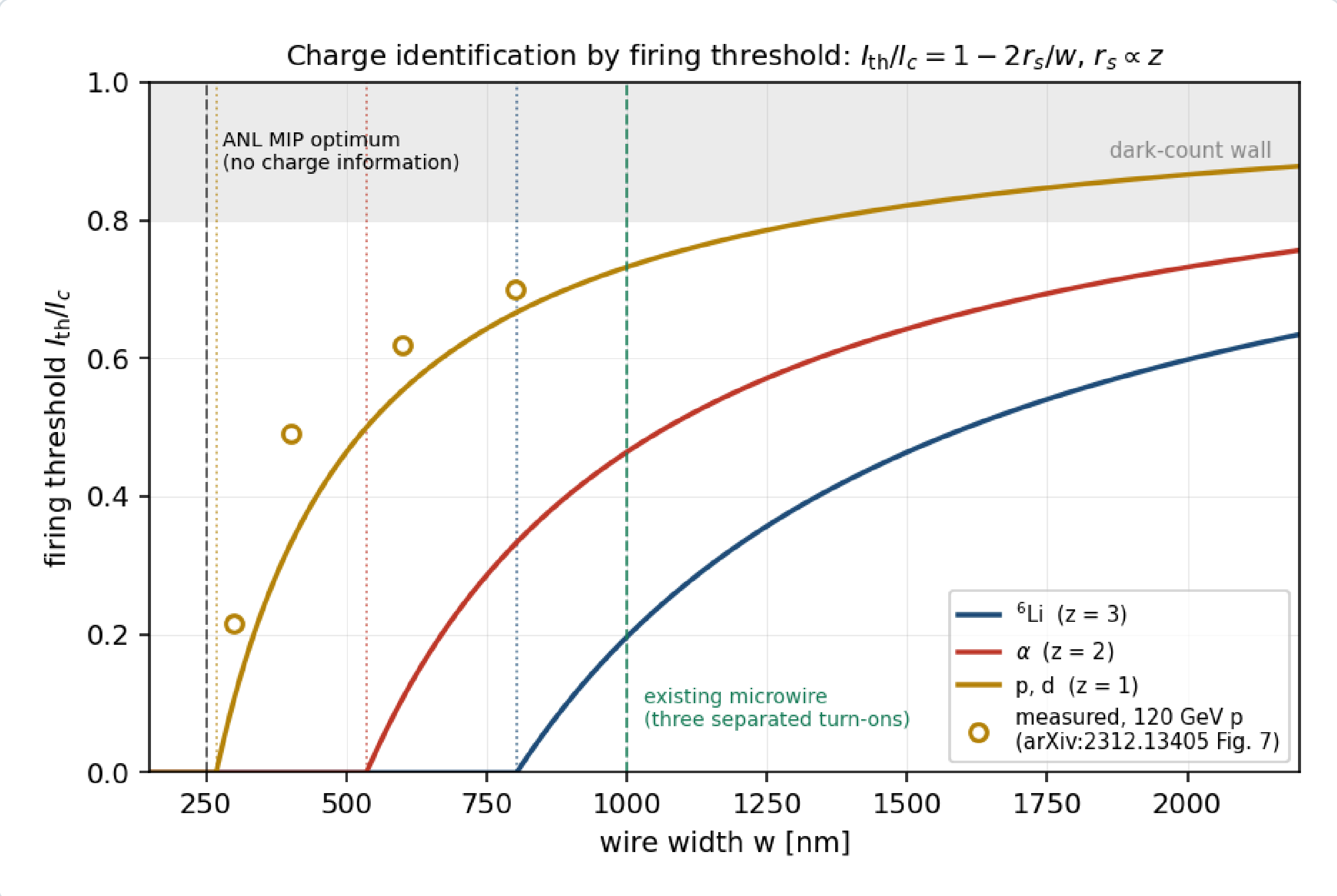

Figure 3 · Z by firing threshold. I_{th}/I_c against wire width for the three species, from $r_s \propto z$ anchored on Argonne's extrapolated 134 nm for a 120 GeV proton. Open circles are their measured thresholds versus wire width, digitised from Fig. 7 of [4]. Dotted verticals mark $w = 2r_s$ for each species — below that width the threshold is zero, the wire fires at any bias, and it carries no charge information at all. The measured points sit above the z = 1 curve because inverting each of them for r_s gives 102–118 nm (mean 113) against the 134 nm Argonne quote as their extrapolated "hard" hot spot; every design number here is a *ratio* between species, so the 15% offset moves nothing — re-anchoring on 113 nm leaves 1 µm separating all three below the noise wall (`evgen/tests/test_nearbeam.py`). `nearbeam_sensor_budget.py` .

WIRE WIDTH	I_{TH}/I_C, P/D	A	⁶LI	SEPARATED TURN-ONS
250 nm	0.00	0.00	0.00	**none** — ANL's MIP optimum
400 nm	0.33	0.00	0.00	**none**
800 nm	0.67	0.33	0.00	**α from d**
1000 nm	0.73	0.46	0.20	**all three** — the existing microwire
1500 nm	0.82	0.64	0.46	**all three**

At w = 1000 nm the three turn-ons sit at 0.20, 0.46 and 0.73 I_C. **Two bias points, at 0.33 and 0.60 I_C, tag Z by the firing pattern alone** — a Z = 3 plane that is blind to α and d, and a Z ≥ 2 plane that is blind to d — and both sit well below the 0.80 I_C at which Argonne measure the dark-count rate rising exponentially [4]. The scheme is not speculative in kind: bias-point charge-state discrimination is a granted patent and was demonstrated on singly- versus doubly-charged lysozyme by subtracting two bias points [9].

> **Two things are in its favour. (i) The Z-ID operating point is also the blackbody-immune one.** A 300 K surface radiates ≈ 3×10^{18} photons cm^{-2} s^{-1} above the 0.04 eV SNSPD threshold, so a photon-mode device cannot look at a warm beam pipe at all. Biasing at 0.33 and 0.60 I_C puts the wire below its photon threshold and leaves only the hard-hot-spot particle mode — which is what the scheme wants anyway. **(ii) The area answer and the Z-ID answer are the same device.** Argonne's efficiency optimum, w ≈ 250 nm, is precisely $2r_s$ for a proton: maximal MIP efficiency and *zero* charge information. Z-ID needs deliberately *wide* wires — the microwire geometry Caltech/JPL/Fermilab already fabricate at 1 µm [7].

> **Three things are against it, and together they are decisive. (i) r_s = 134 nm is not a measurement.** It is the extrapolated zero-crossing of a four-point fit, and the authors write *"While the physical validity of this simple model is a question of future work"* [4]. Inverting their four points individually gives 102–118 nm (mean 113). Quote it as a model parameter, not a datum. **(ii) The volume in which Q is counted is unresolved by a factor of 5000.** The 2024 paper defines Q as *"the energy that the proton has deposited into the thin film"* — ≈ 20 eV. The 2026 paper, with overlapping authorship, applies the same √Q scaling to the *substrate* deposit — 0.1 MeV for the same proton [5]. Both scale as z^2, so $r_s \propto z$ survives; the absolute radii do not. **(iii) "An interpolation, not an extrapolation" is only half true.** Argonne's ^{241}Am α differs from their 120 GeV proton almost entirely through $1/\beta^2$ (β = 0.054 against ≈ 1), not through z^2. Their α point tests the √Q law across a wide energy range; it says nothing about z^2 scaling *at fixed velocity*. **Nobody has ever varied Z at fixed β on one of these devices**, and every published ion demonstration — He^+, lysozyme, the protein beams, the Ar ions — varies kinetic energy through an acceleration voltage, so charge enters as E = zeV, linear in z rather than z^2 from dE/dx.

4.1 · The incumbent carries more information — but far less decisively than a σ suggests

EICROC provides, per channel, a slow shaper path into an **8-bit 40 MHz successive-approximation ADC** for charge alongside its 10-bit 25 ps time-of-arrival TDC — the ToT of ALTIROC was replaced by that ADC specifically to handle dynamic range. Behind it sits an AC-LGAD with a **30 μm active thickness** at 500 μm pitch [11].

> **A σ is the wrong figure of merit, and this note said so with one.** Gap over the quadrature sum of two Landau core widths is neither a PID separation power nor a fake rate. What puts an α inside a ^{6}Li's acceptance window is the Landau *upper tail*, which falls as 1/λ — nothing like a Gaussian. Restated as a fake rate at a matched efficiency, with the Landau sampled rather than approximated (`nearbeam_zid_power.py` , 4 planes, 95% ^{6}Li efficiency, 1.5×10^6 events, 6.7×10^{-7} statistical floor):

READOUT	Α FAKE RATE	EFFICIENCY REACHED
8-bit charge, per-plane likelihood ratio — **the Neyman-Pearson optimum**	2.3×10^{-5}	0.950
one bit per plane, majority-of-*k*, 99% efficient planes	**3.1×10^{-5}**	0.950
truncated mean — the standard analogue dE/dx estimator	2.7×10^{-3}	0.950
plain sum of the four planes	5.3×10^{-2}	0.950

One bit costs a factor 1.4, not orders of magnitude. The species are far apart (MPV 31.7 against 75.2 keV) and the discriminating power comes from requiring *coincidence* across planes, not from precision within one. Note also that a truncated mean is **86× worse than one bit**, and a plain sum worse still: a Landau has no mean, so a single δ ray drags it. More bits does not automatically mean better Z-ID.

4.1a · Where the nanowire actually loses: fill factor

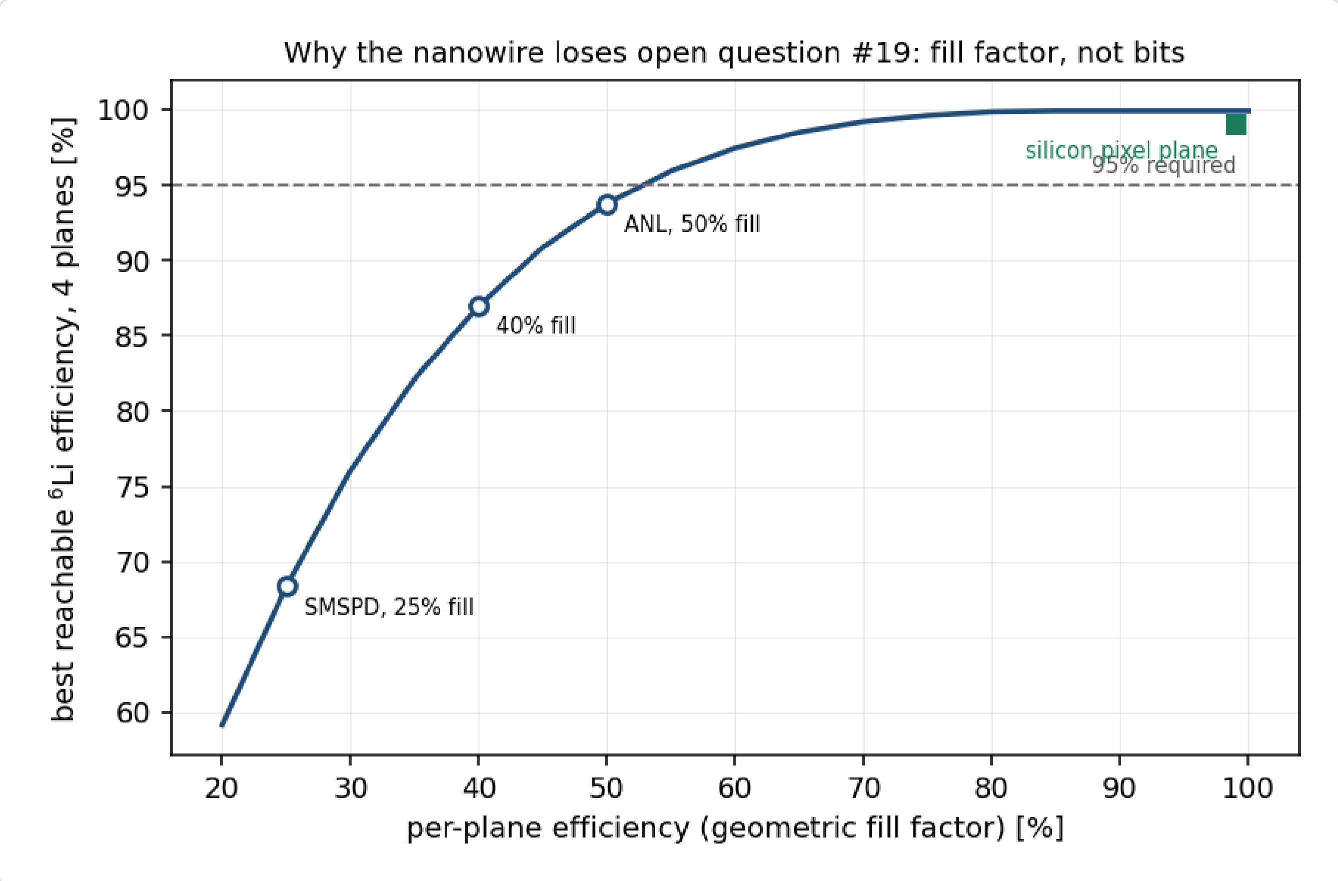

Figure 4 · Fill factor, not bits. The coincidence that makes one bit sufficient needs every plane to record the track. A silicon pixel plane does so ~99% of the time; a superconducting wire comb only over its geometric fill — 25% (SMSPD, [7]), 40% ([8]), 50% (the ANL EIC-targeted device, [4]). At 50% a four-plane array tops out at 93.7% ^{6}Li efficiency and *cannot reach 95% at any working point.* `nearbeam_zid_power.py`.

This is a **fabrication** number, not an information-theoretic one — which makes it a fairer and more actionable thing to put to the MEP group than "you only have one bit", and it is the one item on this list they could straightforwardly change.

The unknown that outranks all of it. EICROC's 8-bit full-scale range is not published anywhere we could reach. A ^{6}Li deposits 75 keV = 3.34 fC unamplified, **≈67 fC at LGAD gain 20** (α: 28 fC), while EICROC0 is documented as "sensitive to low charge (2 fC)" and characterised at 12.5 fC. *If the pot front end is MIP-ranged, the ^{6}Li clips, α and ^{6}Li pile up at full scale, and the incumbent's Z-ID — the load-bearing rejection in this whole programme — is gone.* Compounding it, LGAD gain suppression for highly ionising particles is measured and reaches ~×2.5, compressing α and ^{6}Li toward each other in *collected* charge. That is §7 D4, and it should outrank every sensor comparison in this note.

4.2 · The honest limit on any dE/dx scheme

The limit is the **δ-ray upper tail**, and it is the same for every technology: it falls as 1/λ and is therefore nearly independent of sample thickness. Thickening does not rescue it; independent planes do, because an α needs four upward fluctuations rather than one. That is why the four-plane numbers above sit at 10^{-5} while any single plane is at the percent level — and why the *estimator* matters more than the bit depth.

One caveat runs the other way and makes every fake rate here conservative: the Landau is sampled untruncated, but a δ ray above ~50 keV escapes a 30 μm layer and never deposits. The more tail-sensitive the estimator, the more its quoted fake rate is an overestimate.

4.3 · The handle that is better than either, and free

The background #19 exists to reject is ^{6}Li → α + d. Both fragments sit at beam rigidity, so neither is separated by dispersion — but the breakup's relative momentum is *transverse* and is therefore not boosted. The α at $4p_u$ and the d at $2p_u$

take opposite kicks of the same k_T.

Two momentum scales, and this note first mixed them. $\sqrt{(2\mu Q)}$ with μ = 1247.7 MeV gives **60.7 MeV/c** for Q = the α+d separation energy 1.4743 MeV — the *bound-state* scale κ, which is what `spectator.sample_k` samples — and **42.2 MeV/c** for Q = 0.712 MeV, the decay momentum of the 2.186 MeV 3^+ resonance. Both are physical, for different production mechanisms. The table is now quoted from the repository's own density rather than a single k; the earlier 6.7 / 18.4 / 44.8 mm used k = 40 MeV/c and were ≈1.6× low.

OPTICS	MEDIAN SEPARATION	16–84%	IN 500 MM PIXELS
18 × 275	**10.9 mm**	4.9–22.7	22
10 × 100	**30.1 mm**	13.4–62.3	60
5 × 41	**73.4 mm**	32.6–152	147

R_{12} = 30.6 m, measured at 18 × 275; the other rows carry that lever arm for want of the per-optics value (§7 D3), so read them as scaling.

But the merge is not the dominant one-hit topology. At 5 × 41 the median separation exceeds the pot half-height, so the commoner failure is the partner flying *past the outer edge* or into the beam hole: single-fragment loss is of order **0.9 / 8.8 / 27% per breakup**, against a merge probability bounded by 0.26 / 0.035 / 0.006%. Neither pitch nor momentum resolution recovers a fragment that was never on the sensor. A merged pair, when it happens, deposits Σz^2 = 5 against 9 and the charge readout separates it at ≈4σ per plane wherever it sits.

An intact ^{6}Li is one hit; the breakup is two, tens of pixels apart, in sensors that already exist. Topology is a stronger discriminant than dE/dx for exactly the background #19 was written about, and nobody in this programme had looked at it.

5 · What stands in the way

None of the following is a physics objection. All of them are engineering or programmatic, and all of them are load-bearing.

OBSTACLE	THE NUMBER	STATUS
Cold-stage power budget	A NbN device needs 2.8–3 K, a WSi microwire 0.8 K. SNSPD cold-stage loads are hundreds of µW to a few mW. A 4 K pulse tube delivers 1.5 W at 4.2 K and *zero* at its 2.8 K no-load floor, on a 190 kg water-cooled compressor drawing 10.7 kW.	**hard**
Beam-induced RF heating	A ferrite-shielded TOTEM Roman Pot dissipates ≈ 40 W at the LHC. Scaling the resistive-wall/broadband factor $Q^2M/\sigma_z^{3/2}$ to EIC parameters gives ≈ 16 / 8 / 0.8 W at highest-ECM / highest-L / 41 GeV. Even the 41 GeV case exceeds a pulse tube's entire 4.2 K lift.	**hard**
In-vacuum precedent	None found anywhere for a biased superconducting detector inside an accelerator's *beam* vacuum. The precedents — LHC CryoBLM silicon and diamond at 1.9 K, the CERN AD SQUID current comparator — sit outside it, and the AD beam is ~10^7 antiprotons against the EIC's 8×10^{13} protons.	**open**
Vibration	Cryocooler cold-head displacements are specified at 7–9 µm; a purpose-built decoupling structure brings that to 10 nm but is heavy and stiff — the opposite of a pot that rides motorised rails to within 10σ of the beam every fill.	**mitigable**
Radiation hardness	Unmeasured. The FY22 proposal's central deliverable was *"an upper limit for the accumulated dose ... This currently unknown limit"*; four years on, no publication reports the LEAF irradiation, the JLab Hall C parasitic run or the SEU cross-section. The 2023 Quantum Sensors for HEP report says SNSPDs *"are expected to be radiation hard ... Although published studies in this area are lacking"*. The whole "radiation hard, therefore closer" argument is an anticipation — the proposal itself writes "(anticipated)".	**unmeasured**
Area	The largest SNSPD/SMSPD ever run in a GeV particle beam is 2 × 2 mm² with 8 channels [8]. A near-beam strip 3 mm deep over 20 mm of vertical extent, both sides of the beam, is 120 mm² — 30× the state of the art, and 1000× if the whole 9.3 mm gap is instrumented. Tiled as 30 × 30 µm² nanowire devices that is 133,000 channels; as 1 × 1 mm² microwire tiles it is 960.	**funded R&D**

Latching	Latching is a hard failure, not a dead time: the device locks resistive and stops detecting. Argonne's ^{241}Am α run is the reassurance — a ≈ 1 μm hot spot did *not* latch a 100–200 nm wire, and gave clean count-rate plateaus [5].	**addressed**
Rate	*Not* a blocker. The ePIC Preliminary Design Report puts pot rates at a few Hz per channel in normal running and *"30–50 kHz at ~10σ"* from beam halo, on STAR experience — four orders below the ~10^8 counts/s a nanowire's fall time allows [6].	**closed**

Cryogenics also inverts the packaging argument. The nanowire's advantages — windowless operation in the beam vacuum, an active edge a few hundred nanometres from the substrate rim, no vacuum-window material — are real and are exactly what a near-beam sensor wants. But a device that cannot see a 300 K surface must be enclosed in a cold shield, and a cold shield is material in front of the sensor. The windowless argument and the blackbody constraint pull against each other, and the resolution — bias below the photon threshold, §4(ii) — is available but has never been demonstrated next to a beam.

6 · A baseline correction: the ePIC pot geometry has moved

Independent of nanowires, and more immediately actionable. The aperture this repository measured was taken in the September-2024 `epic-main` inside `jug_xl-nightly`. Reading the current `main` branch of `eic/epic` directly [3]:

	SEPTEMBER-2024 (WHAT `TOOLS/FULLSIM` MEASURED)	CURRENT `MAIN`
module	`32 × 32 mm`	`16 × 16 mm`
insertion	energy-*independent*: 3.2 cm outer, +0.7 cm central	per-energy 10σ offsets in `beamline_*.xml`
RF shield	1 mm aluminium on each module face, *active*	commented out — *"we don't know if we will even need it ... Oct. 2025"*
implied blind block at 18 × 275	`32 mm (x) × 7 mm (y)`	`16 mm (x) × 2.7 mm (y)`

The old geometry's 32 mm block gives 32 mm / 30.6 m = 1.046 mrad against the 1.03 mrad the scan measured — agreement to 1.5%, which is the check that the measurement and the file reading confirm each other.

The correction runs in opposite directions at the two ends, which is why it cannot be guessed. At 18 × 275 the current 16 mm block implies ≈ 0.52 mrad — *below* the 0.727 mrad 10σ envelope, so the sensor package no longer binds and the beam does. At 5 × 41 the per-energy insertion moves the other way, to a 29.6 mm inner edge, because the pots now retract properly for the larger low-energy beam — so the aperture there gets *worse*, not better. That is precisely the configuration at which the coherent programme was said to survive.

> **What this means for the numbers in circulation.** The measured aperture, the sign-flipped ⟨cos 2β⟩ = −0.77, the $1.4{\times}10^{-2}$ tagged fraction, and the conclusion that *"the coherent programme is a low-energy programme"* are all conditioned on a superseded geometry. They should be re-measured against the current release before being quoted further. The conversion from angle to millimetres also needs the per-optics R_{12}, which was measured for 18 × 275 alone (30.6 m); this note therefore works in angle throughout and quotes millimetres only for that configuration. **The curves of Figure 1 are unaffected** — they price every aperture, which is why they are plotted that way; what moves is only the marker labelled "you are here".

7 · What to ask, and of whom — cheapest first

#	QUESTION	OWNER	WHY IT DECIDES SOMETHING
D1	**Re-measure the pot aperture on the current `epic-main`**, per optics, with the RF shields as they now stand.	this repository (plans/09 B1)	**The highest ratio of conclusion-change to effort anywhere here.** One container run. Every aperture-conditional number depends on it, and the correction runs in opposite directions at the two ends (§6).

D2	What actually sets the 10σ retraction at IP6 — sensor survival, or machine protection and the collimation hierarchy? And what is σ_x at z = 32.5 m for the *light-ion* optics?	C-AD / ePIC FF WG	A conversation, not an experiment. The first part decides whether any "rad-hard ⇒ closer" argument can work at all. The second is worth three orders of magnitude in coherent yield at top energy by itself: the repository's envelope implies $10\sigma_x \approx 22$ mm while Jentsch's "1σ ~ 1 mm" implies ~10 mm.
D3	Per-optics R_{12} and R_{34}.	ePIC FF WG	R_{12} = 30.6 m was measured for 18 × 275 only, which is why this note works in angle and quotes millimetres for that optics alone.
D4	**Settle EICROC's input charge dynamic range in fC, and LGAD gain suppression at ~9 MIP.** A ^{6}Li deposits ≈67 fC at gain 20; the chip is characterised at 12.5 fC. Then: can the existing AC-LGAD stack do Z-ID? A Geant4 study through all four layers with an EICROC front-end model, plus two numbers: the chip's **input charge dynamic range in fC**, and the sensor's **gain-suppression curve** at ~9 MIP.	ePIC FF WG / OMEGA-IJCLab / BNL AC-LGAD group	**If the pot front end is MIP-ranged the ^{6}Li clips and the incumbent's Z-ID is gone** — a far larger risk to the programme than any sensor comparison here. Then one person-month of Geant4 closes #19 either way (§4.1). Ask this before anything about nanowires.
D5	Two-hit topology for α + d: what fraction of breakup events put both fragments in acceptance, and how well does multiplicity separate them from an intact ^{6}Li?	this repository	Free, already instrumented, 13–90 pixels of separation, and unexamined (§4.3).
D6	The x-moving layer and a staircase retraction near x = 0.	ePIC FF WG (A. Jentsch)	Where the ×60–×330 actually lives, and already being designed — *"Expect to have this finalized in the next 6-8 weeks"* (July 2025).
D7	*Only if D4 says the LGAD cannot do it:* the α-versus-proton turn-on bias on the same wires.	ANL MEP	Costs them one plot from data they already hold, and kills the concept at zero cost if the α curve is not where √Q predicts. Be precise that it tests the √Q law, **not** z^2 at fixed β — the experiment that would test that is a relativistic light-ion beam on a 1–2 μm microwire, and this study should not be what motivates it.

8 · Conclusion

There is no design here in which superconducting nanowires measurably improve far-forward lithium tagging, and both candidate roles fail on arithmetic rather than on unknowns. The near-beam role fails because the sensor's dead edge is worth 0.005 mrad against a gap of 0.5–3 mrad set by module tiling and the 10σ rule, and because the 10σ rule protects the machine rather than the sensor. The charge-ID role fails because the incumbent AC-LGAD already digitises an 8-bit analogue charge over a 30 μm active layer — strictly more information per plane than a nanowire's one bit — across four planes ePIC already has, and because the limit on Z separation is the δ-ray tail, which is the same for both.

That is a conclusion about *this* application, not about the technology. The Argonne programme's own four-application list points elsewhere — the cold bore of a superconducting magnet, in front of the ZDC, a Compton polarimeter — and the field result that makes the technology unique (saturated efficiency to 5 T parallel to the device plane [10]) is worth nothing at a pot in a field-free drift and everything inside a magnet.

What this study leaves behind is worth more than its negative answer. Figure 1 prices any closer approach by any means, and says the mid-energy configuration is worth ×569 and the top energy nothing. §6 says the geometry those numbers were anchored to no longer exists. And §4.3 says the background that motivated open question #19 is separable by hit topology in sensors that are already baselined. The ranked list of what would actually help the lithium tags is: the low-energy configuration; re-tiling the near-beam silicon, which ePIC is already designing; using the charge EICROC already digitises; two-hit topology for the α + d channel; and the IR-8 secondary focus. Superconducting nanowires appear nowhere on it.

References

[1] R. Abdul Khalek et al., *Science Requirements and Detector Concepts for the Electron-Ion Collider: EIC Yellow Report*, Nucl. Phys. A 1026 (2022) 122447, arXiv:2103.05419, §14.5.3 (`refs/2103.05419_part4.pdf`): the four EIC SNSPD applications, of which the first is the far-forward Roman Pot; edgeless operation "within a few 100 nm of the substrate edge"; membranes "as thin as few 10 µm"; the mm² pixel-array goal.

[2] W. Armstrong (PI) et al., *Superconducting Nanowire Detectors for the EIC*, FY22 EIC-Related Generic Detector R&D proposal #18, §3.3 Concept 1 — reviewed and partially funded (Report of the 2022 EIC-related Generic Detector R&D Review Committee, June 2023). The 10σ rule-of-thumb passage quoted in §3.

[3] `eic/epic`, branch `main`: `compact/far_forward/roman_pots_eRD24_design.xml` and `compact/fields/beamline_{18x275,10x100,5x41}.xml`, read directly (2026-08-26). Module 1.6 × 1.6 cm; per-energy offsets with the comment "These are the ten-sigma cuts for the Roman pots, translated to the physical layout we currently have. They are not perfectly ten-sigma for reasons of physical geometry."; RF shield module components commented out, "Oct. 2025".

[4] S. Lee, T. Polakovic, W. Armstrong, A. Dibos, T. Draher, N. Pastika, Z.-E. Meziani, V. Novosad, *Beam Tests of SNSPDs with 120 GeV Protons*, Nucl. Instrum. Meth. A 1069 (2024) 169956, arXiv:2312.13405 (`refs/2312.13405.pdf`). 12 nm NbN, 30 × 30 µm², wire widths 300–800 nm at 2.82 K; Eqs. 1–2; r_S = 134 nm; ~250 nm optimum; background rising exponentially above I_b/I_c = 0.8.

[5] S. Lee, W. Armstrong, J. DiPreta, C. Dulya, V. Novosad, T. Polakovic, *Optimization of Cryogenic Detector Test Station by Rejecting Electromagnetic Interference*, Nucl. Instrum. Meth. A 1093 (2027) 171953, arXiv:2601.03158 (`refs/2601.03158.pdf`). The 5.5 MeV ^{241}Am α at ≈ 1 µm hot spot by the same √Q scaling; clean plateaus with no latching; "a detailed study of α detection is underway".

[6] ePIC Collaboration, *Preliminary Design Report*, 29 September 2024, the Roman Pots and off-momentum detectors section: "Rates during normal operations, with expected vacuum of 10^{-9} mbar, are a few Hz/channel. However, the beam halo could potentially provide rates of 30–50 kHz at ~10σ from experience of Roman pots at STAR."

[7] C. Wang et al. (Caltech / JPL / Fermilab), *Towards High-Efficiency Particle Detection Using Superconducting Microwire Arrays*, arXiv:2510.11725 (`refs/2510.11725.pdf`). 8-channel 1 × 1 mm² WSi, 4.7 nm film, 1 µm wires on a 3 µm gap (25% fill), 0.8 K; 75% fill-factor-normalised efficiency and 130 ± 17 ps at CERN SPS H6.

[8] C. Peña, C. Wang, S. Xie et al., *Characterization of a Superconducting Microwire Single-Photon Detector Array for Charged-Particle Detection*, JINST 20 (2025) P03001, arXiv:2410.00251 (`refs/2410.00251.pdf`). 2 × 2 mm², 8 channels, 30 ± 1 µm spatial resolution; amplitude distributions indistinguishable across species; sensor edge sharper than the 10 µm telescope.

[9] R. Cristiano, M. Ejrnaes, A. Casaburi, N. Zen, M. Ohkubo, *Superconducting nano-strip particle detectors*, Supercond. Sci. Technol. 28 (2015) 124004: the hot-spot threshold in the ion literature; the charge-state discrimination demonstrated on lysozyme by bias-point subtraction (Suzuki et al., Rapid Commun. Mass Spectrom. 24 (2010) 3290; US Patent 8,872,109 B2, Ohkubo & Suzuki, AIST). Also J. J. Renema et al., arXiv:1301.3337, for the photon-side form of the same relation.

[10] T. Polakovic, W. R. Armstrong, V. Yefremenko, J. E. Pearson, K. Hafidi, G. Karapetrov, Z.-E. Meziani, V. Novosad, *Superconducting nanowires as high-rate photon detectors in strong magnetic fields*, Nucl. Instrum. Meth. A 959 (2020) 163543, arXiv:1907.13059 (`refs/1907.13059.pdf`). Saturated efficiency to 5 T *parallel* to the device plane but only ≈ 0.5 T perpendicular; 10^7 counts/s measured, ~10^8 /s from τ_F = 11.78 ns.

Appendix A · Revision history

DATE	REVISION
2026-08-27	**The charge argument is corrected, and it was overstated.** "4.8σ per plane" is neither a PID separation power nor a fake rate; restated as a sampled-Landau fake rate at matched efficiency, **one bit per plane costs a factor 1.4 against the Neyman-Pearson optimum**, and a truncated mean costs a factor 86 *to* one bit (§4.1, Figure 4, `nearbeam_zid_power.py`). The nanowire loses on **geometric fill factor** — 50% fill cannot reach 95% ^{6}Li efficiency over four planes — not on bit depth. Also corrected: the α+d separation mixed the bound-state scale κ = 60.7 MeV/c with the 3^+ resonance decay momentum 42.2 MeV/c, and is now quoted from the repository's own density (median 10.9 / 30.1 / 73.4 mm, ≈1.6× the earlier single-k figures, §4.3); the dominant one-hit topology is single-fragment *loss* (0.9 / 8.8 / 27% per breakup), not merging (≤0.26%); and EICROC's unpublished dynamic range is promoted to the top of §7.
2026-08-26	First version, then revised the same day against an adversarial review. **The verdict changed sign.** As first written, this note concluded that charge identification was the strong half of the case; the review established that the incumbent AC-LGAD already digitises an 8-bit charge over a 30 µm active layer across four planes, giving ≈ 4.8σ of α ↔ ^{6}Li separation per plane against a nanowire's one bit (§4.1) — so a nanowire is a downgrade, not an upgrade, and open question #19 was being asked of the wrong technology. Three further corrections: r_S = 134 nm is an *extrapolated* model parameter and not a measurement; the volume in which Q is counted differs by a factor 5000 between Argonne's own two papers; and their ^{241}Am α point tests the √Q law across β, not z² at fixed velocity, so the "interpolation" argument is only half true (§4). The dead-edge argument of §3 was retired as worth 0.005 mrad against a 0.5–3 mrad gap, the granularity argument redirected to the x-moving layer ePIC is already designing, and §4.3 added — the α + d breakup separates by 13 to 90 pixels of the existing 500 µm pitch, which is a better answer to #19 than any dE/dx scheme and is free. The original content: The aperture scan and the coherent chain at both apertures (Figures 1–2, `nearbeam_aperture_scan.py`, `nearbeam_reach_gain.py`, and `--near-beam-mrad` added to

`money_cos2phi_coherent_reco.py`); the hot-spot firing-threshold model of charge identification (Figure 3, `nearbeam_sensor_budget.py`); the obstacle table; the ePIC geometry correction of §6, read from the current `main` branch; the question list of §7.

All acceptances use B = 50 GeV^{-2} for the coherent |t| slope and the high-acceptance optics (σ_θ = 72.7 μrad, 10σ = 0.727 mrad). The measured silicon aperture is from `tools/fullsim` in the September-2024 `epic-main` — see §6. Energy deposits are Bethe means in a 12 nm NbN film; the Landau most-probable value is deliberately not quoted, because ξ/I ≈ 0.002 in this film puts it far outside the ξ ≫ I regime the formalism requires. Hot-spot radii are anchored on Argonne's *extrapolated* 134 nm — a fit parameter of a model whose authors write that "the physical validity of this simple model is a question of future work" — and scaled as z. Built by `reports/build_report.py`; figures from the `evgen/scripts` named in the text; display mathematics typeset at build time.